



# Prepared For Anything

The day of the standalone imaging solution is over,  
and the time is ripe to spring for a new multifunctional device

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MFDs, also known as all-in-ones are slowly winning over customers. Versatility differentiates MFDs from other peripheral products, and is a major reason why we think MFDs make more sense than stand-alone products such as printers or scanners.

If we analyse some activities we do on our home computer, it becomes very clear that printing is not our only requirement—scanning and copying are also much needed features. Now that we're using the Internet for booking tickets online, paying bills, getting a statement of accounts and so on, printouts are needed. That's not all—digitising docu-

ments such as bills, policies, legal documents, medical reports, and what have you, require a scanner, and then there's always the need to photocopy a document or two at short notice. Thus, as we move forward in an Internet enabled world, the need for having a MFD is more relevant than it was a couple of years ago.

While MFDs have been around for years now, they haven't been able to make any impact, largely due to technical and economic reasons. On the technical front, earlier MFDs couldn't perform as well as their stand-alone counter parts and hence didn't catch the fancy of consumers. However, the same printing and scanning engines used on stand-alone products are making their way

into MFDs, so performance is no longer an issue. They print fast, produce excellent scans, deliver more than acceptable copies and some can be used for fax as well. With inkjet technology getting better, some MFDs deliver photo-quality printouts and have compliance with technologies such as PictBridge or come with memory card readers for direct printing from digital cameras. In recent years, prices have fallen to an extent that MFDs are proving more cost-effective than stand-alone devices.

Similar to Printers, MFDs are also available in both Inkjet as well as Laser engines. Most Inkjet MFDs fall in the entry level bracket and offer good value for money; however they don't offer

top of the line performance. Laser MFDs are generally expensive, but have slight speed advantage over inkjets and are cost-effective if your printing volume is high. In this test we take an extensive look at inkjet MFDs and also have chosen few Laser MFDs which fit a home user's budget.

We categorised the products in three main categories. The first category consists of products priced less than Rs 7,000. People on a shoe string budget should look at this category. The second category consists of products priced between Rs 7,000 and 12,000. People who want better features should look at this category. The third category consists of products priced between Rs 12,000 and Rs 18,000. These products have all cutting edge features and are meant for technology savvy people.

## Entry Level MFDs

In this category we have rounded up products that offer bare minimum multi-function features. Canon's PIXMA MX-308, Brothers DCP-350, EPSON's CX5500 and CX7300 made into this category.

## Features

Though MFDs are peripheral devices, they definitely can be designed to look good and Brother has done every effort to impress us with their DCP-350. It is compact, has pearl white finish and sports clean design. Other MFDs from Canon and Epson are boxy, relatively less compact and have unappealing aesthetics.

Canon's MX308 was the only MFD here to have fax capability, but it is also the priciest of them. If you want faxing too then the MX308 offers standard faxing features and offers storing of 50 fax pages in the device runs out of paper.

While Canon's MX308 uses the PC's memory, the Brother DCP-350 is equipped with 32 MB onboard input buffer. Both Epson CX-5500 and CX-7300 have paltry 64 KB of onboard buffer and it shows up in the performance. None of these devices



have automatic duplex printing; however it can be manually done from the driver setting page.

Brother's DCP-350 and Epson's CX-7300 were the two devices to be equipped with both a PictBridge port and memory card reader enabling

direct printing from digital cameras. Canon's MX-308 supports PictBridge but lacks memory card reader while Epson's CX-5500 doesn't have any. All devices here had a maximum paper capacity of 100 sheets; good enough for home usage.

## How We Tested

### The Test Rig

Our test bed for the inkjet MFD tests comprised an Intel Pentium 4 560 3.2 GHz processor, a Gigabyte GA-965P-DQ6 motherboard with 1 GB of Micron 533 MHz DDR2 RAM, an NVIDIA GeForce 7600GT graphics card, and a 120 GB 7200 rpm ATA IV Seagate Barracuda hard disk. We used Windows XP with SP2 as the OS.

### Media Used

Text/combi-document printouts: 100 gsm paper from Berga (A4 size)  
Photo printouts: 160 gsm glossy photo paper from Novajet (A4 size)

### Features

Various features such as PictBridge, memory card readers, fax, input tray capacity, number of cartridge, etc were noted. For the scanner part, the maximum scan resolution, bit depth, scanner type, ADF presence, etc. were noted and rated. For the copier aspect, the ability to work as a standalone device, the maximum number of pages that can be printed in multi-copy mode, reduction or enlarging a copy, and so on, were noted.

### The Printer Tests

We used a text document to test the raw

print speeds of the printer part of the MFDs. To test the ability of the MFDs to handle the various aspects of a regular document, we created a combi document with black text interspersed with graphs and images. The quality settings used were Normal and Best for text and combi document, and Best for photo. We clocked the time taken in each case, and analysed the print quality.

### The Scanner Tests

We tested the scanners for speed using an average of five previews of an A4 size image. This was done with the scanner fresh out of the box, in order to incorporate the warm-up time. We imported the same A4 size image at 150 dpi and 600 dpi in Adobe Photoshop CS2, and scanned a full text document in B&W mode at 200 dpi. The scanner then underwent a test to determine its colour differentiating abilities using a Kodak IT8 card with different shades of colour—each a little different from the preceding one.

### The Copier Tests

We copied the text document in black and white, and the combi-document in colour, and rated the MFDs on their speed and print quality.